Auteur: Siebe Mees
Studierichting: Informatica
Oefeningenreeks 5I nr. 2.2

$f(x) = \ln(x^2 - 1)$ op $[2, 5]$

x	2		5
$f(x)$	$\ln(3)$	$+$	$\ln(24)$
$f'(x)$		$+$	
$f''(x)$		$-$	
$f(x)$	$\ln(3)$	$\nearrow$	$\ln(24)$
		$\frown$	

`5I-2.2-siebe.mees.INF.png`

$$f'(x) = \frac{2x}{x^2 - 1}$$

$$L = \int_2^5 \sqrt{1 + (f'(x))^2} dx$$

$$= \int_2^5 \sqrt{1 + \left(\frac{2x}{x^2 - 1}\right)^2} dx$$

$$= \int_2^5 \sqrt{1 + \frac{4x^2}{(x^2 - 1)^2}} dx$$

$$= \int_2^5 \sqrt{\frac{(x^2 - 1)^2 + 4x^2}{(x^2 - 1)^2}} dx$$

$$= \int_2^5 \frac{\sqrt{x^4 - 2x^2 + 1 + 4x^2}}{x^2 - 1} dx$$

$$= \int_2^5 \frac{\sqrt{x^4 + 2x^2 + 1}}{x^2 - 1} dx$$

$$= \int_2^5 \frac{\sqrt{(x^2 + 1)^2}}{x^2 - 1} dx$$

$$= \int_2^5 \frac{x^2 + 1}{x^2 - 1} dx$$

$$
\begin{aligned}
&= \int_2^5 \left(1 + \frac{2}{x^2 - 1}\right) dx \\
&= \int_2^5 \left(1 + \frac{1}{x - 1} - \frac{1}{x + 1}\right) dx \\
&= [x + \ln(x - 1) - \ln(x + 1)]_2^5 \\
&= (5 + \ln 4 - \ln 6) - (2 + \ln 1 - \ln 3) \\
&= 3 + \ln\left(\frac{4}{6} \cdot \frac{3}{1}\right) \\
&= 3 + \ln 2
\end{aligned}
$$

References